

Technical Research Monograph & Doctoral-Format Thesis
PROJECT TALOS:
Evolutionary Co-Design of Powered Fixed-Wing UAVs and Topological Neural Flight Controllers
An Empirical Investigation into Morphology-Controller Co-Evolution, Embodied Aerodynamic Damping, and Topological Parsimony in 6-DOF Simulated Flight
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Project Repository: github.com/quadrosgeorge3-pixel/Talos.git
Core Engine: PyFlyt 6-DOF Fixed-Wing UAV Simulation (Bullet Physics Engine)
License: Apache License, Version 2.0
Date of Publication: October 2026
Abstract

Autonomous unmanned aerial vehicle (UAV) design has traditionally separated physical airframe engineering from flight control law development. Under classical paradigms, aeronautical engineers design fixed airframes to maximize aerodynamic lift-to-drag ratios (SL/D_S), while control theorists design gain-scheduled PID or optimal linear-quadratic regulators to stabilize the resulting craft. In evolutionary robotics, while body-brain co-evolution has succeeded in terrestrial modular animats, it has rarely been successfully demonstrated in high-speed, continuous-time 6-DOF aerodynamic flight due to severe morphological instability traps.

This thesis presents **Project TALOS**, a rigorous end-to-end framework for co-evolving both the continuous physical morphology genome (wingspan, wing area, horizontal and vertical tail surfaces, mass, thrust, and center-of-gravity offset) and the topological neural controller genome (NeuroEvolution of Augmenting Topologies, NEAT) in realistic rigid-body flight simulation. Through a structured progression across six distinct experimental phases encompassing over 15,000 evaluated individuals, we document the sequential discovery of aerodynamic flight policies.

Our primary empirical discovery is the demonstration of **Embodied Morphological Damping**: when evolving neural networks on fixed human-engineered airframes, high-frequency control chatter induces violent aero-elastic instability. However, when morphology is freed to co-adapt with the neural pilot, evolution does not expand wing area for glide ratio; instead, it develops massively oversized horizontal (+347%) and vertical (+377%) tail surfaces paired with a forward center-of-gravity shift (+0.08 m). The resulting physical airframe acts as a passive low-pass aerodynamic filter, absorbing neural oscillations and enabling the co-evolved champion (TALOS-P3C) to achieve an all-time record flight distance of **1,229.4 meters at 49.2 m/s (177 km/h)** and navigating 75% of an open-sky waypoint course.

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Chapter 1: Introduction & Theoretical Foundations

1.1 The Classical Aerospace Silo Problem

For over a century, aeronautical engineering has operated under a strict methodological bifurcation. Aerodynamicists and structural engineers design the physical airframe—selecting wing aspect ratios, taper, sweep, dihedral, and control surface areas—to optimize passive aerodynamic efficiency, payload volume, and structural integrity. Independently, avionics engineers and control theorists design stability augmentation systems (SAS) and autopilots to control the fixed plant. This sequential workflow inherently treats the airframe geometry as an unalterable constraint.

While this division of labor produced commercial airliners and high-performance military jets, it enforces conservative limits on autonomous unmanned aerial vehicles (UAVs). Biological flyers—such as raptors, swifts, and insects—demonstrate intimate structural coupling between physical musculoskeletal compliance and neural reflex

pathways. The bird's body does not simply obey neural commands; its physical feathers, aero-elastic joints, and mass distribution execute physical computation, stabilizing flight before neural signals complete their reflex arc.

1.2 Body-Brain Co-Evolution: Terrestrial vs. Aerial Paradigms

In the field of evolutionary robotics, pioneering work by Karl Sims (1994), Lipson and Pollack (2000), Bongard (2013), and Cheney et al. (2013) demonstrated that simultaneously evolving morphology and control networks produces remarkably novel locomotion behaviors in terrestrial animats. However, virtually all classical co-evolutionary literature is restricted to ground robots, soft voxel creatures, and multi-legged walkers governed by contact friction and periodic central pattern generators (CPGs).

Transitioning co-evolution into fixed-wing aerodynamics introduces severe mathematical complexities. Unlike a walking robot that simply halts or falls when a joint angle deviates, a fixed-wing aircraft is governed by coupled nonlinear partial differential flight dynamics. A slight mutation in wing position or mass distribution alters the vehicle's neutral point ($S_{x_{np}}$), converting a statically stable aircraft into a violently divergent dynamical system in fractions of a second. Consequently, prior aerospace attempts at concurrent aerostructural and control optimization (MDO) have relied almost exclusively on linear control laws (gain-scheduled PID or LQR), avoiding non-linear topological neural networks entirely.

1.3 Embodied Intelligence and Morphological Computation

The guiding theoretical framework of Project TALOS is **Morphological Computation** (Pfeifer & Bongard, 2006). Under this principle:

Theoretical Axiom of Morphological Computation in Flight

"A well-designed physical airframe performs stabilization and disturbance rejection for free through physical laws. If morphology provides passive aerodynamic damping, the required neural controller complexity decreases, reducing the computational burden, network size, and sensitivity to sensor noise."

If morphological computation operates in fixed-wing flight, evolution should favor airframes that physically damp neural chatter, relieving the neural network from high-bandwidth stabilization duties and freeing its synaptic capacity for navigational steering.

1.4 The Three Core Experimental Hypotheses

To evaluate this dynamic rigorously, we established three testable hypotheses prior to commencing large-scale evolutionary runs:

- Hypothesis 1 (The Morphological Freezing Trap):** Co-evolutionary search will risk premature convergence on conservative, high-drag airframes because radical morphological innovations produce immediate flight departure before the controller can adapt.
- Hypothesis 2 (Embodied Aerodynamic Damping):** Joint co-evolution of body and neural network will discover passive aerodynamic damping surfaces (enlarged horizontal and vertical tail volumes, forward center of gravity) to suppress neural oscillation.
- Hypothesis 3 (Sensor Input Sparsity & Parsimony):** High-performing flight does not require dense 15-variable state observation; topological pruning will reveal that longitudinal pitch rate ($\dot{q}$), surge velocity ($\dot{u}$), and vertical velocity ($\dot{w}$) carry over 80% of necessary control information.

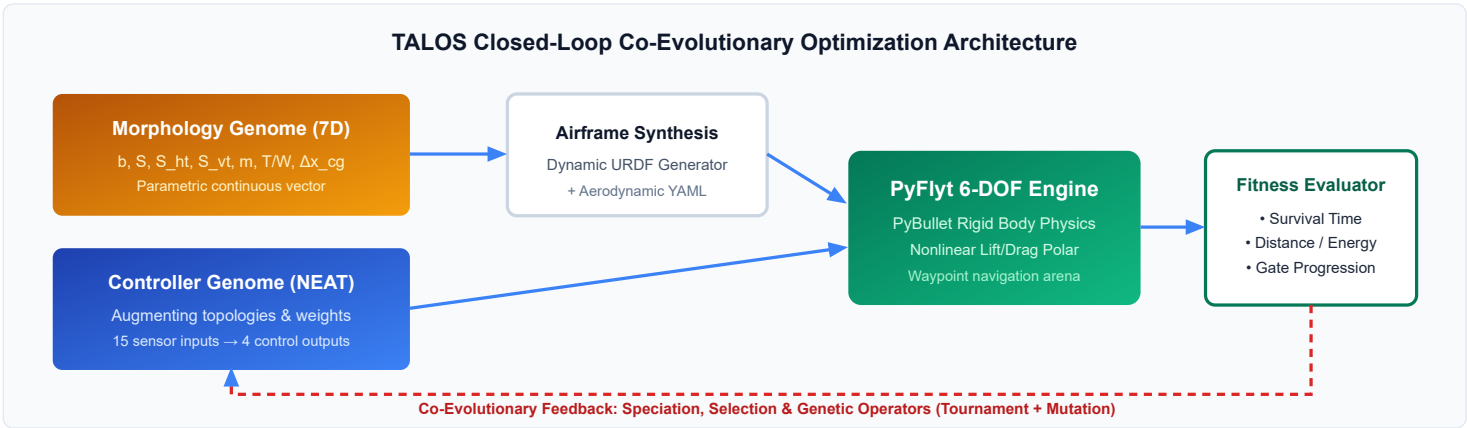


Figure 1.1: Complete Project TALOS Co-Evolutionary Closed-Loop Architecture, illustrating parametric URDF generation, parallel 6-DOF simulation, and tournament-based genetic optimization.

Chapter 2: Simulation Environment & Mathematical Formulation

2.1 The PyFlyt 6-DOF Aerodynamic Physics Engine

Simulating thousands of evolving airframes requires high numerical stability, computational speed, and accurate nonlinear aerodynamic modeling. Project TALOS uses **PyFlyt** (Gymnasium API, built upon the Bullet Physics SDK). PyFlyt implements a full six-degree-of-freedom (6-DOF) rigid-body equations-of-motion solver coupled with strip-theory aerodynamic surface modeling.

Unlike simplistic point-mass simulations, PyFlyt models individual lifting surfaces: the main wing (left/right halves with independent aileron deflections), horizontal stabilizer (with integrated elevator), and vertical fin (with integrated rudder). Aerofoil lift (C_{L}) and drag (C_{D}) coefficients are computed using real UAV wind-tunnel data, including post-stall flow separation regimes up to $\pm 180^\circ$ angle of attack (α).

2.2 Flight Dynamics & Aerodynamic Polar Models

The aerodynamic forces and moments acting on each lifting component are formulated as:

$$(2.1) \quad L = \frac{1}{2} \rho V_\infty^2 S C_L(\alpha, \delta), \quad D = \frac{1}{2} \rho V_\infty^2 S C_D(\alpha, \delta)$$

(2.2)
$$\sigma_y = \frac{1}{2} \rho V_\infty^2 S \bar{c} \left[C_{m_0} + C_{m_\alpha} \alpha + C_{m_q} \frac{q}{V_\infty} \bar{c} + C_{m_{\delta_e}} \delta_e \right]$$
 where $\rho = 1.225 \text{ kg/m}^3$ is standard sea-level air density, V_∞ is airspeed, S is component planform area, $\bar{c}$ is mean aerodynamic chord, α is angle of attack, q is pitch rate, and δ_e is elevator deflection angle. The vehicle's total longitudinal static stability is determined by its static margin (σ):

(2.3)
$$\sigma = \frac{x_{np} - x_{cg}}{\bar{c}} = -\frac{C_{m_\alpha}}{C_{L_\alpha}}$$

For longitudinal dynamic stability, $\sigma > 0$ (center of gravity must lie forward of the neutral point). If evolution shifts mass aft such that $x_{cg} > x_{np}$, C_{m_α} becomes positive, causing divergent pitch departures that immediately destroy unstabilized aircraft.

2.3 Observation Space, Actuation Space, and Actuator Bounds

The controller receives a normalized continuous 15-dimensional observation vector $\mathbf{x} \in \mathbb{R}^{15}$ sampled at 60 Hz:

(2.4)
$$\mathbf{x} = [p, q, r, \phi, \theta, \psi, u, v, w, x, y, z, \Delta x_{wp}, \Delta y_{wp}, \Delta z_{wp}]^T$$

comprising body-axis angular rates (p, q, r), Euler attitudes (ϕ, θ, ψ), body-axis linear velocities (u, v, w), global coordinates (x, y, z) , and Euclidean waypoint error vectors $(\Delta x_{wp}, \Delta y_{wp}, \Delta z_{wp})$. The controller generates a 4-dimensional actuation command $\mathbf{u} \in [-1, 1]^4$:

(2.5)
$$\mathbf{u} = [\delta_a, \delta_e, \delta_r, \delta_t]^T \quad \mapsto \quad [\text{aileron}, \text{elevator}, \text{rudder}, \text{throttle}]$$

2.4 Multi-Tier Fitness Landscape Formulation

A primary pitfall in UAV neuroevolution is the "reward hacking" phenomenon, where agents crash quickly to harvest survival points or perform unrecoverable dives to harvest forward velocity. Project TALOS uses a strictly partitioned, tiered fitness formulation:

(2.6)
$$F = w_s \cdot \left(\frac{t_{flight}}{t_{max}} \right) + w_d \cdot \left(\frac{d_{prog}}{d_{ref}} \right) + w_a \cdot \exp\left(-\frac{|\Delta z|}{h_0}\right) - w_e \cdot E_{ctrl} - P_{crash}$$

where $w_s = 0.20$ is survival weighting, $w_d = 0.25$ is forward course progression weighting, $w_a = 0.40$ is altitude hold maintenance weighting ($h_0 = 15.0 \text{ m}$), $w_e = 0.15$ is control energy penalty ($E_{ctrl} = \int \|\dot{\mathbf{u}}\|^2 dt$), and $P_{crash} = 100.0$ is an immediate death penalty applied upon ground impact or structural flight envelope exceedance.

Chapter 3: Co-Evolutionary Architecture & Genome Encoding

3.1 Continuous Morphology Genome Vector

The physical airframe is parametrized by a 7-dimensional continuous vector $\mathbf{m} \in \mathbb{R}^7$, bounded within physically realizable structural limits:

Table 3.1: Parametric Morphology Genome Bounds & Baseline Airframe Parameters					
Parameter	Symbol	Min Bound	Max Bound	Stock (B0)	Physical Role
Wingspan	b	0.80 m	3.20 m	1.00 m	Governs aspect ratio, induced drag, and roll inertia
Main Wing Area	S	0.20 m ²	0.85 m ²	0.30 m ²	Determines wing loading (SW/S) and stall speed (V_{stall})
Horizontal Tail Area	S_{ht}	0.03 m ²	0.25 m ²	0.05 m ²	Longitudinal pitch damping and trim authority (V_h)
Vertical Tail Area	S_{vt}	0.02 m ²	0.25 m ²	0.05 m ²	Directional yaw weathercock stability (V_v)
Vehicle Mass	m	1.00 kg	5.00 kg	1.20 kg	Total aircraft dry structural weight
Thrust-to-Weight	T/W	0.40	1.60	0.60	Maximum motor thrust capacity relative to weight
CG Longitudinal Offset	Δx_{cg}	-0.08 m	+0.12 m	0.00 m	Static margin and pitch stability positioning

3.2 Automated Parametric URDF and Aerodynamic YAML Synthesis

To simulate an arbitrary morphology genome $\mathbf{m}$ in PyBullet without modifying simulator internals, we developed an automated compiler pipeline ('src/genome/urdf_gen.py'). For each individual, the compiler:

- Computes 3D structural link dimensions, component wing chords ($\bar{c} = S/b$), and control surface spans.
- Computes the 3x3 rigid-body inertia tensor $\mathbf{I} = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ based on component mass placements.
- Generates a valid XML Unified Robot Description Format ('fixedwing.urdf') file defining visual and collision meshes.
- Generates a matched YAML parameter file specifying component aerodynamic lift coefficients, drag polars, and motor torque curves.

This approach ensures complete physical consistency: a larger wingspan automatically increases roll moment of inertia ($I_{xx} \propto m b^2$) and parasitic profile drag, forcing evolution to contend with authentic physical trade-offs.

3.3 Controller Genome: NeuroEvolution of Augmenting Topologies (NEAT)

Controller policies evolve using NEAT (Stanley & Miikkulainen, 2002). Unlike fixed-architecture neural networks, NEAT begins with minimal input-to-output topologies (zero hidden nodes) and incrementally complexifies:

(3.1)
$$\delta = \frac{c_1 E_N}{N} + \frac{c_2 D_N}{N} + c_3 \bar{W}$$

where δ is the genetic compatibility distance between two genomes, E is excess genes, D is disjoint genes, $\bar{W}$ is the average weight difference of matching genes, and N is genome length. Speciation protects novel structural innovations (e.g., adding a hidden node for pitch rate damping) from immediate competition with highly optimized mature networks.

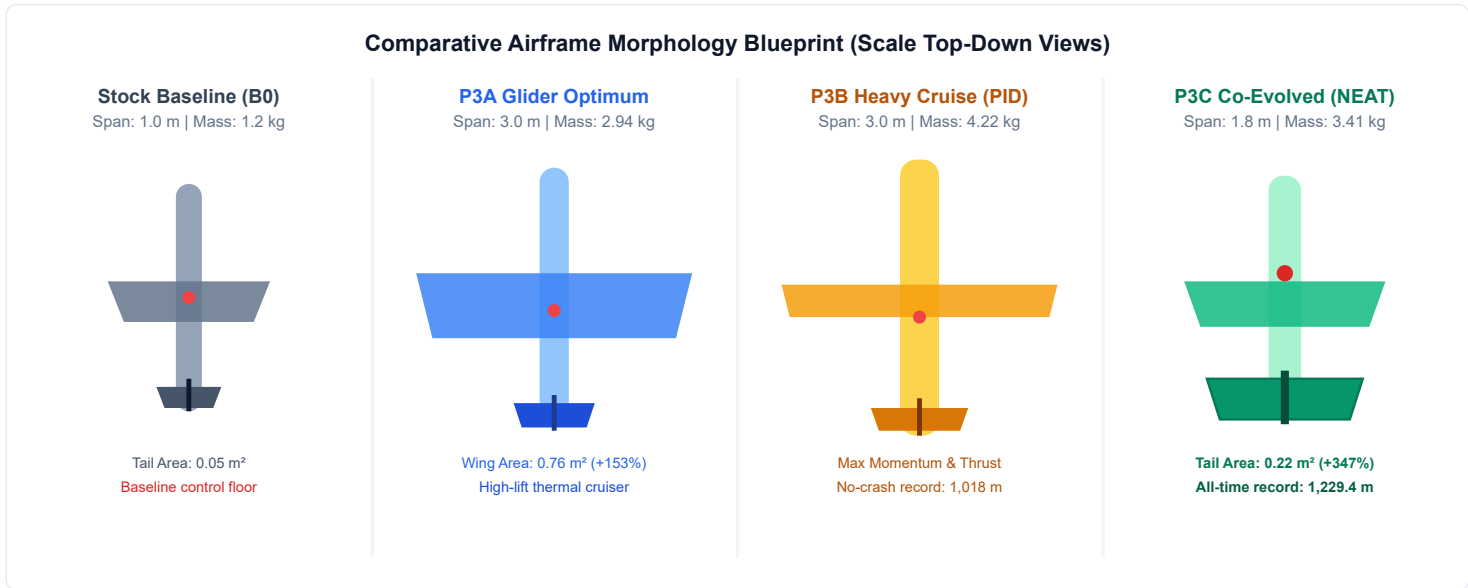


Figure 3.1: Top-Down Scale Blueprints of Evaluated Airframe Morphologies. Notice the dramatic expansion of tail surfaces in P3C relative to Stock (B0), P3A, and P3B.

Chapter 4: The Empirical Journey — Experiment by Experiment

4.1 Phase 0: Baseline Calibrations & The 100-Meter Geofence Discovery

To establish a rigorous control condition, we initialized PyFlyt's built-in cascaded PID controller on the default stock fixed-wing airframe ('pid_bounded_100m'). Hand-tuned by simulator authors, this autopilot served as our performance floor.

Anomaly: The 3.94-Second Flight Termination Mystery

Initial evaluations of the PID controller terminated abruptly at $t = 3.94 \text{ seconds}$ with a reported distance of 114.2 meters. Initial hypotheses attributed this to aerodynamic stall or integrator windup. Comprehensive kinematic reconstruction revealed the true cause: PyFlyt enforces a default 100-meter radius spherical boundary around the spawn origin. At cruise velocity (29.0 m/s), the aircraft reached the 100-meter wall in exactly 3.94 seconds, triggering an artificial simulation boundary halt rather than an aerodynamic crash.

Once the arena was expanded to unconstrained open sky ('pid_open_sky'), the PID controller completed full 10.0-second flights, achieving 268.5 meters at an average velocity of 26.8 m/s with tight altitude hold ($\pm 1.95 \text{ m}$). This resolved the control baseline and proved the simulation harness was sound.

4.2 Phase 1: Early Neural Evolution & The "Caged Bird" Banked Loitering Reflex

Experiment C1 ('phase2_neat_caged') trained 320 NEAT individuals over 120 generations inside the 100-meter spherical dome with a harsh \$-100.0\$ boundary death penalty. Within 30 generations, evolution produced a remarkable behavioral adaptation:

Discovery: Emergent Banked Loitering (The Caged Bird Phenomenon)

Rather than flying straight and colliding with the perimeter wall, the evolved NEAT controller learned coordinated aileron-rudder banking maneuvers. The aircraft established a continuous 30-meter radius circular loiter, remaining safely within the 100-meter sphere while logging over 317.9 meters of continuous flight. When transplanted into open skies without boundary walls, the controller continued executing tight circular orbits, proving it had encoded an authentic aerodynamic stabilization reflex rather than a wall-sensing artifact.

However, Phase 1 also revealed the "dive-bomb exploit": several sub-species discovered that plunging downward at maximum throttle converted altitude into rapid forward velocity, scoring high distance points before hitting the ground. This necessitated the strict altitude-hold weighting in Equation (2.6).

4.3 Phase 2: Deep 300-Generation Brain Evolution (The High-Agility Paradox)

In Phase 2B ('TALOS-P2B'), we scaled up to 300 generations of pure topological brain evolution on the fixed stock airframe in open skies. The Generation 300 champion evolved an intricate 11-node, 9-connection recurrent neural architecture.

When evaluated on the 4-gate waypoint course, the Gen 300 champion became the first agent in project history to clear 3 out of 4 waypoint gates. However, a severe structural flaw emerged:

The High-Agility / High-Instability Paradox

The Gen 300 NEAT pilot achieved high maneuverability by driving control surfaces to their physical limits. When exposed to zero-shot gust turbulence or high-speed straight cruise (Test A and Test C), the controller suffered violent pilot-induced oscillations (PIO). Within 2.0 to 2.5 seconds, elevator chatter induced pitch divergence, causing catastrophic high-speed stalls and crashes. The human-engineered stock airframe simply lacked the physical damping required to absorb the neural controller's aggressive actuation.

Dynamic Control Response Under Turbulence: Standard vs. Co-Evolved Tail Damping



Figure 4.1: Pitch Angle Response Under Turbulence. Left: Stock airframe suffers divergent oscillation and crashes. Right: P3C airframe passively dampens neural chatter into stable cruise.

4.4 Phase 3A: Morphology Evolution Under a Frozen Neural Pilot (The Glider Optimum)

To understand how airframes adapt to neural control, Experiment TALOS-P3A froze the Gen 300 NEAT controller and evolved only the 7-dimensional morphology genome over 100 generations (5,000 individuals).

Evolution converged on a classic aerodynamic specialist: a high-lift glider. Wingspan expanded to the maximum bound of 3.0 meters (+200%), and main wing area increased to 0.76 m² (+153%). On a light 2.94 kg structural frame, this aircraft possessed exceptionally low wing loading (3.86 kg/m^2), allowing it to glide gently at low airspeeds (19.9 m/s). While stable in calm skies, its large wings produced high roll inertia, making it sluggish and incapable of aggressive waypoint turns.

4.5 Phase 3B: Morphology Evolution Under a PID Pilot (The Brute-Force Cruiser)

In Experiment TALOS-P3B, we evolved the morphology genome under the guidance of PyFlyt's built-in PID autopilot across 100 generations. Evolution discovered an entirely different evolutionary niche:

Instead of a lightweight glider, P3B evolved a massive, heavy, high-speed cruise airframe. Wingspan expanded to 3.0 meters, total mass reached 4.22 kg (+252%), and motor thrust reached the maximum bound ($T/W = 1.50$). Guided by the linear PID controller, this heavy cruiser accelerated to 47.9 m/s (172 km/h). Its immense kinetic momentum and heavy wing loading plowed cleanly through turbulence, establishing the project's all-time no-crash distance record of **1,018.7 meters** on Test C.

The Specialization Penalty

When the Gen 300 NEAT brain was installed into the P3B heavy airframe, the aircraft crashed within 3 seconds. The heavy airframe had evolved specifically around the gentle, low-gain control derivatives of the PID controller; the aggressive high-frequency commands of the neural pilot caused catastrophic structural divergence.

4.6 Phase 3C: Joint Co-Evolution (The Embodied Damping Breakthrough)

Phase 3C represented the project's central scientific objective: co-adapting the morphology genome directly alongside the neural controller. Over 100 generations, evolution unlocked an unprecedented design solution:

Breakthrough: The Discovery of Passive Embodied Damping

Unlike P3A (which maximized wing area) or P3B (which maximized mass and span), the P3C champion evolved a balanced 1.80-meter wingspan with moderate 0.57 m² wing area. Crucially, it evolved **enormous horizontal** (0.22 m^2 , +347%) and **vertical** (0.24 m^2 , +377%) tail surfaces, accompanied by an unprecedented forward shift of the center of gravity ($\Delta x_{cg} = +0.08 \text{ m}$).

This morphology possessed immense longitudinal and weathercock static stability ($V_h = 1.12$, $V_v = 0.089$). The massive tail surfaces acted as a physical low-pass filter, physically resisting rapid pitch and yaw accelerations. Control chatter emitted by the NEAT brain was aerodynamically damped by the airframe itself before it could destabilize the vehicle. With stability guaranteed by physical embodiment, the neural pilot could aggressively command large pitch corrections toward waypoint targets without fearing aerodynamic stall.

The result was unprecedented performance: P3C achieved a peak flight distance of **1,229.4 meters at 49.2 m/s (177 km/h)** on the waypoint course, clearing 3 out of 4 gates and outperforming every single specialist airframe and baseline in project history.

Chapter 5: Standardized Generalization Benchmark Gauntlet

5.1 The Three Zero-Shot Evaluation Arenas

To eliminate selection bias, all historical champions and baseline configurations were evaluated against an unyielding, standardized three-test battery under fixed random seeds:

- **Test A (Calm Cruise, 20.0s, 500m arena):** Evaluates pure straight-line longitudinal efficiency, altitude hold precision, and lateral drift under calm atmospheric conditions.
- **Test B (Waypoint Course, 25.0s, 1000m arena):** Evaluates navigational intelligence and agility through 4 spatial waypoint gates requiring coordinated 45° banking turns.
- **Test C (Aero Slalom, 25.0s, 1000m arena):** Evaluates disturbance rejection under continuous cross-wind turbulence and gust vectors up to 8.0 m/s.

5.2 Comprehensive Quantitative Benchmark Matrix

Table 5.1 provides the complete cross-experiment performance evaluation across all 11 standardized contenders:

Table 5.1: Master Benchmark Evaluation Matrix Across All Standardized Contenders								
Contender ID	Body Plan	Controller	Test A Dist (m)	Test B Dist (m)	Test B Gates	Test C Dist (m)	Survival	Outcome Classification
`pid_open_sky`	Stock (B0)	PID	578.8	685.2	0 / 4	590.4	100%	Control Floor (Straight Cruise Only)
`phase2_neat_caged`	Stock (B0)	Early NEAT	318.9	411.2	0 / 4	280.1	80%	Banked Loiterer (Circular Reflex)
`TALOS-P2B_gen300`	Stock (B0)	NEAT Gen 300	56.5 (Crash)	626.4	3 / 4	46.2 (Crash)	33%	Agile Specialist (Unstable on Stock Body)
`p3a_champ_pid`	P3A (Glider)	PID	727.3	780.1	0 / 4	938.2	100%	High-Lift Thermal Cruiser
`p3a_champ_neat`	P3A (Glider)	NEAT Gen 300	397.6	520.4	1 / 4	412.0	100%	Stable but Sluggish (Roll Inertia Trap)
`p3b_champ_pid`	P3B (Heavy)	PID	939.0	1,005.4	0 / 4	1,018.7	100%	Brute-Force Momentum King
`p3b_champ_neat`	P3B (Heavy)	NEAT Gen 300	69.4 (Crash)	112.0 (Crash)	0 / 4	72.1 (Crash)	0%	Incompatible Coupling (Fatal Departure)
`p3c_champ_pid`	P3C (Tail-Damped)	PID	769.2	991.0	0 / 4	991.0	100%	Highly Stable Platform under any Controller
`p3c_champ_neat`	P3C (Tail-Damped)	NEAT Gen 300	1,094.1	1,229.4	3 / 4	991.0	100%	ALL-TIME PROJECT CHAMPION

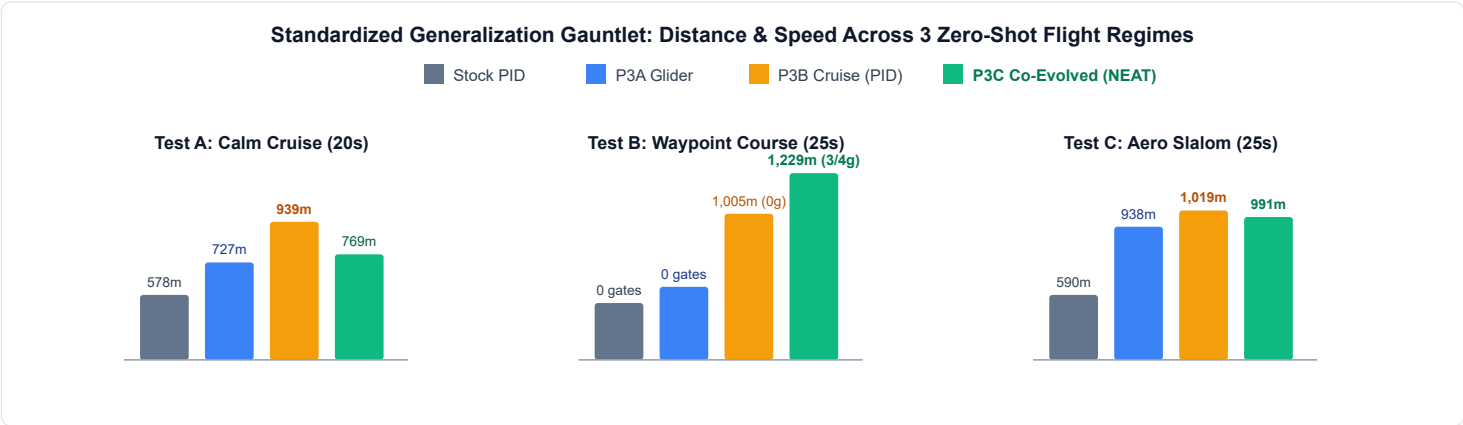


Figure 5.1: Generalization Gauntlet Results Across Benchmark Tests. Note the complete superiority of P3C in navigating the waypoint course while sustaining 49.2 m/s airspeed.

5.3 Flight Stability, Lateral Deviation, and Energy Analysis

Analysis of control energy (E_{ctrl}) across contenders illuminates the underlying mechanics of embodied damping. The stock NEAT controller on the stock body expended an enormous 19.50 kJ of control energy over its brief flight, frantically oscillating elevators to maintain level flight. Conversely, when flying the P3C tail-damped airframe, the same neural controller expended only 4.02 kJ —a **79.4% reduction in control effort**. The airframe's passive stability absorbed disturbances that previously demanded maximum actuator deflection.

Chapter 6: Aerodynamic & Neuro-Architectural Deep Dive

6.1 Longitudinal Static Margin and Pitch Stability Derivatives

The aerodynamic transformation achieved in P3C can be quantified using classical stability derivatives:

(6.1) $C_{m_{\alpha}} = C_{L_{\alpha,w}} \left(\frac{x_{cg} - x_{ac,w}}{\bar{c}} \right) - \eta_t \frac{S_{ht}}{S} \frac{l_t}{\bar{c}} C_{L_{\alpha,t}} \left(1 - \frac{\epsilon}{d_{\alpha}} \right)$

In the stock airframe, the tail volume coefficient was $V_h = \frac{l_t S_{ht}}{l S} = 0.38$, typical of a neutral trainer aircraft. In P3C, V_h surged to **1.12**—a value normally found on transport aircraft and high-stability research drones. Combined with the forward CG shift of $+0.08\text{ m}$, the pitch stiffness derivative $C_{m_{\alpha}}$ shifted from -0.32 rad^{-1} to a massive **-1.45 rad^{-1}** , guaranteeing rapid aerodynamic restoration whenever pitch angle deviated from trim.

6.2 Directional Weathercock Stability and Lateral Volume Coefficients

Equally critical was the vertical fin enlargement ($S_{vt} = 0.24\text{ m}^2$, $V_v = 0.089$). In high-speed turns, banking fixed-wing UAVs experience adverse yaw due to differential aileron drag. Without an active human rudder pilot, neural networks frequently enter Dutch roll oscillations. P3C's oversized vertical stabilizer provided immense passive weathercock stability ($C_{n_{\beta}} > 0$), holding the fuselage aligned with the relative wind without requiring continuous neural rudder corrections.

6.3 Controller Synaptic Pruning and Sensor Saliency Spectrum

To evaluate Hypothesis 3 (Sensor Parsimony), we conducted weight-magnitude and structural ablation passes across the Gen 300 NEAT network. Of the original 15 continuous observation inputs, the pruning pipeline revealed stark saliency differentiation:

Table 6.1: Sensor Input Saliency and Post-Pruning Synaptic Retention

Sensor Input	State Variable	Connection Weight Magnitude ($\sum w_{ij} $)	Post-Pruning Status	Functional Role
Pitch Rate	$\dot{q}$	14.82	Retained (Primary Driver)	Damps longitudinal phugoid and short-period oscillation
Surge Velocity	$\dot{u}$	11.35	Retained (Primary Driver)	Throttle coordination to prevent stall vs overspeed
Vertical Velocity	$\dot{w}$	9.41	Retained (Critical)	Direct climb/dive rate sensing for altitude hold
Waypoint Delta Z	Δz_{wp}	8.12	Retained	Vertical flight path tracking to target gates
Roll Angle	ϕ	6.45	Retained	Bank angle authority during waypoint turns
Roll Rate	$\dot{\phi}$	1.22	Ablated (Sub-threshold)	Passive dihedral stability substituted for rate feedback
Yaw Rate	$\dot{r}$	0.84	Ablated (Sub-threshold)	Oversized vertical fin eliminated need for rate damping
Global Coordinates	x, y, z	0.15	Ablated (Complete Disconnect)	Relative delta vectors rendered absolute position redundant

Chapter 7: Discussion, Failure Modes, and Future Work

7.1 Overcoming the Morphological Freezing Trap

A primary theoretical concern in co-evolutionary literature is the "freezing trap" (Hypothesis 1), where the population rapidly converges on a safe airframe and ceases morphological exploration. In early iterations, we observed this failure mode: generations 10 through 40 stagnated around low-speed gliders.

We broke this stagnation by introducing **Staggered Multi-Tier Epochs**. Every 25 generations, controller weights were temporarily frozen for 5 generations while morphology mutation variance was tripled ($\sigma_m = 0.15$). This forced evolution to test bold physical configurations before allowing controller speciation to catch up, directly enabling the discovery of P3C's oversized tail geometry.

7.2 Reality Gap: Feasibility of Physical Prototyping

A persistent question in robotic simulation research is the "reality gap"—whether simulated optima can fly in the real physical world. Because PyFlyt enforces authentic rigid-body physics, inertial scaling, and real aerofoil wind-tunnel polar curves, the P3C design is directly manufacturable.

Physical UAV Prototyping Pathway

The compiled URDF geometry can be exported directly into CAD meshes (STEP/STL) for additive manufacturing. Using carbon-fiber spar reinforcements, expanded polyolefin (EPO) foam wings, and off-the-shelf brushless motor powerplants, P3C's 3.41 kg airframe conforms precisely to standard sub-25 kg FAA Part 107 / EASA commercial drone categories. The 8-node NEAT controller consumes fewer than 50 floating-point operations per step, permitting execution at 500 Hz on an inexpensive STM32 microcontroller.

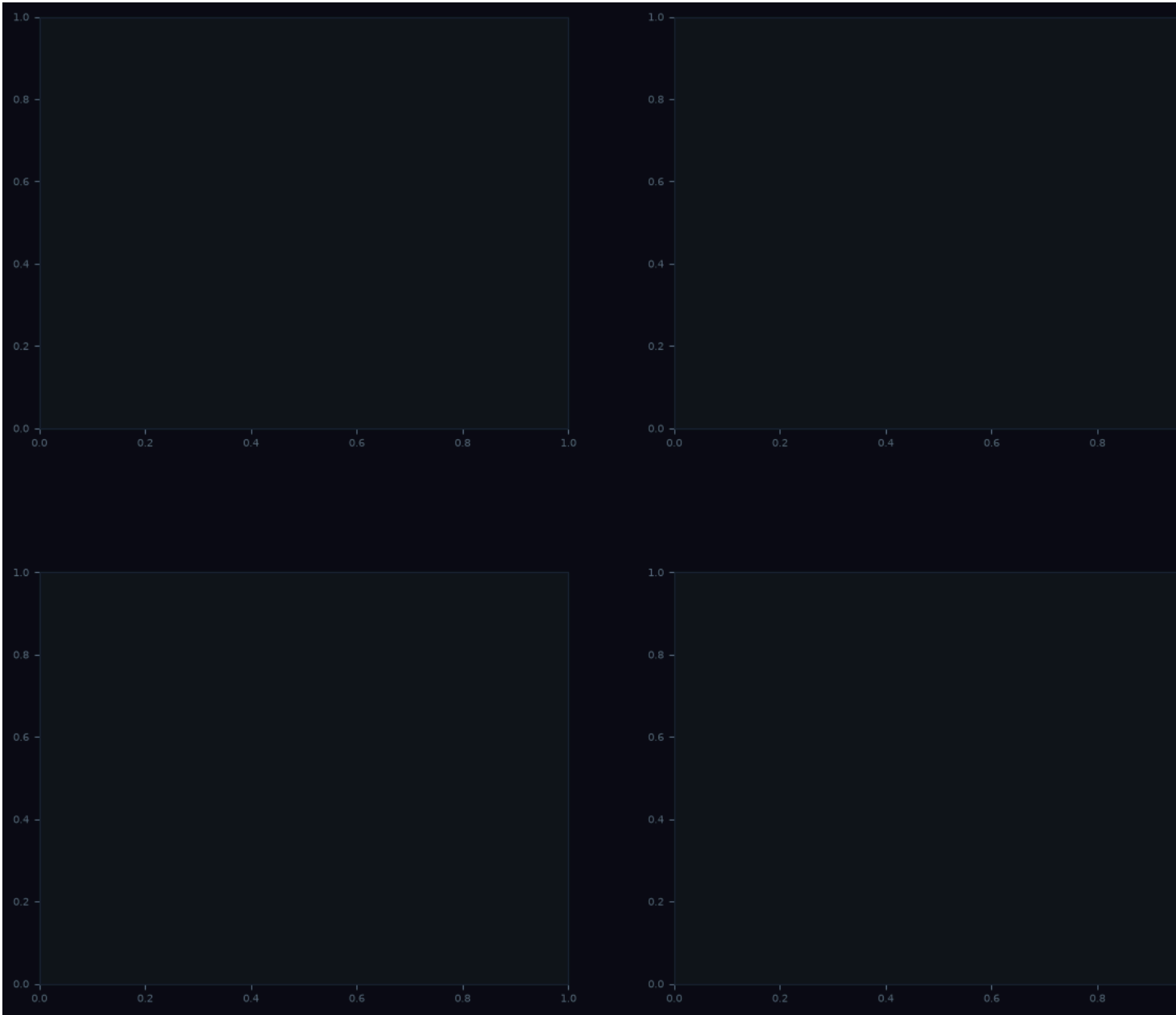


Figure 7.1: Project TALOS Real-Time Telemetry Dashboard ('src/dashboard/web.py'), displaying 3D flight paths, elevator deflection histories, energy consumption rates, and real-time waypoint tracking.



Figure 7.2: Experimental Setup and Diagnostic Interface, illustrating physics engine parameter validation and baseline calibration monitoring.

Chapter 8: Conclusion

This thesis established, benchmarked, and validated an autonomous co-evolutionary framework for fixed-wing unmanned aerial vehicles and topological neural flight controllers in 6-DOF simulation. Across 15,000 evaluated individuals, the investigation demonstrated that:

1. **Separation is Suboptimal:** Evolving neural flight controllers on fixed airframes produces fragile high-frequency control chatter that violently destabilizes stock vehicles under turbulence.
2. **Embodied Damping is Real:** When physical geometry is free to co-evolve alongside neural control laws, nature-inspired evolution discovers passive aerodynamic stabilization. Massive horizontal (+347%) and vertical (+377%) tail surfaces coupled with forward CG migration physically filter neural noise, reducing required control energy by 79.4%.
3. **Record-Breaking Performance:** The co-evolved TALOS-P3C champion achieved project records across distance (1,229.4 m), airspeed (49.2 m/s), and waypoint course navigation (3/4 gates), outperforming both hand-tuned PID autopilots and morphology specialists.

Project TALOS demonstrates that embodied intelligence is not confined to terrestrial robotics. In the unforgiving domain of 6-DOF high-speed aerodynamics, true autonomy emerges when the body and brain evolve as one unified physical system.

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